1) $\begin{bmatrix} 2 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 3 \end{bmatrix} + -2 = 2-3 + -2 = -3$

2) $z = (-0.5)(4) + 1 = -1$

$\sigma(1) = \frac{1}{1+e^{+1}} = \frac{1}{1+e} = 0.27$

3) $\begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$

Relu $\begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$

$\begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \end{bmatrix} + 0 = -3$

4) $\frac{1}{2}(-1-2)^2 = \frac{1}{2}(-3)^2 = \frac{9}{2}$

5) $\frac{\partial L}{\partial \hat{y}} = \hat{y} - y$

6) $\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} + 1 = w_1 x_1 + w_2 x_2 + b_1 = \hat{y} = -1$

$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w_1} = (\hat{y} - y) \cdot x_1 = -1 \cdot 1 = -1$

$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w_2} = \hat{y} - y \cdot x_2 = -1 \cdot -2 = 2$

$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial b} = \hat{y} - y \cdot 1 = -1 \cdot 1 = -1$

7) $w_1 = 0 - 0.2(-1) = 0.2$

$w_2 = 1 - 0.2(2) = 0.6$

$b = 1 - 0.2(-1) = 1.2$

8) $z = wx + b$ $\qquad z = -3$

$-3 = -2 + -1$ $\qquad a = 0$

$L = \frac{1}{2}(-3)^2 = \frac{9}{2}$

$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial w} = 0 \cdot -1 = 0$

$\frac{\partial L}{\partial b} = 0 \cdot 1 = 0$

9) $\hat{y} = 0$

$\frac{1}{2}(\hat{y} - y)^2 = \frac{1}{2}$

$\frac{\partial L}{\partial w} = (-1)(2) = -2$

$\frac{\partial L}{\partial b} = -1 \cdot 1 = -1$

$w_1 = 0 - 0.2(-2) = 0.2$

$b_1 = 0 - 0.2(-1) = 0.1$

$\hat{y} = (0.2)(2) + 0.1 = 0.5$

$\frac{1}{2}(\hat{y} - y)^2 = \frac{1}{2}0.5^2 = 0.125$

$\frac{\partial L}{\partial w} = -0.5 \cdot 2 = -1$

$\frac{\partial L}{\partial b} = -0.5$

$w_2 = 0.2 - 0.1(-1) = 0.3$

$b_2 = 0.1 - (0.1)(-0.5) = 0.15$

(b) $\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{1+e+e^2} \\ \frac{e}{1+e+e^2} \\ \frac{e^2}{1+e+e^2} \end{bmatrix}$

11) In forward pass, you predict using the given parameters. In back-prop, you adjust the parameters given the output from forward-prop

12) The network wouldn't be able to predict anything and limite

13) a) It wouldn't converge
b) It would take to long to converge

14) When gradients become very small and the layers receive values from the previous that are practically 0 which makes prediction diff

15) It reduces vanishing gradients and focuses in positives.